Netlist Extraction

Prof. Jeong Hoan Park

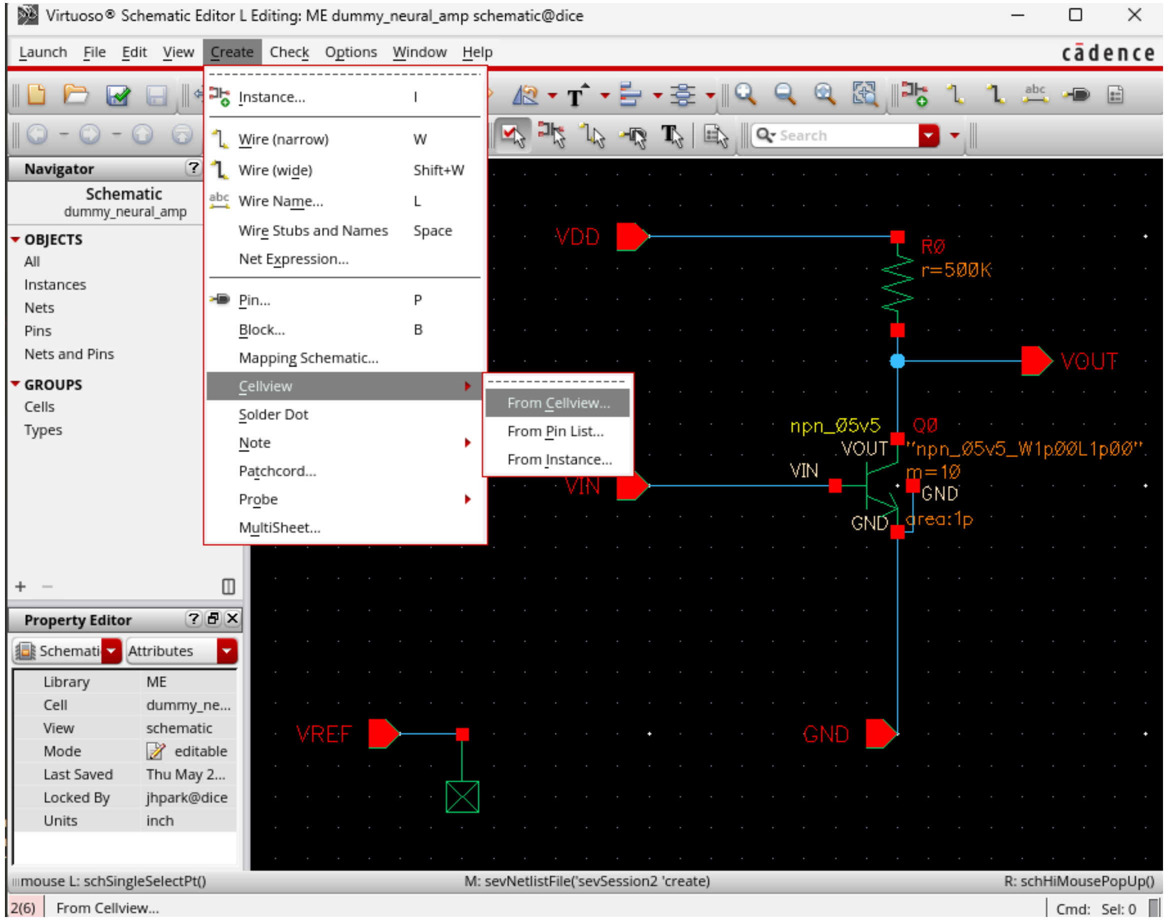
Kyung Hee University

j.h.park@khu.ac.kr

Draw Schematic

❑Schematic Creation Flow

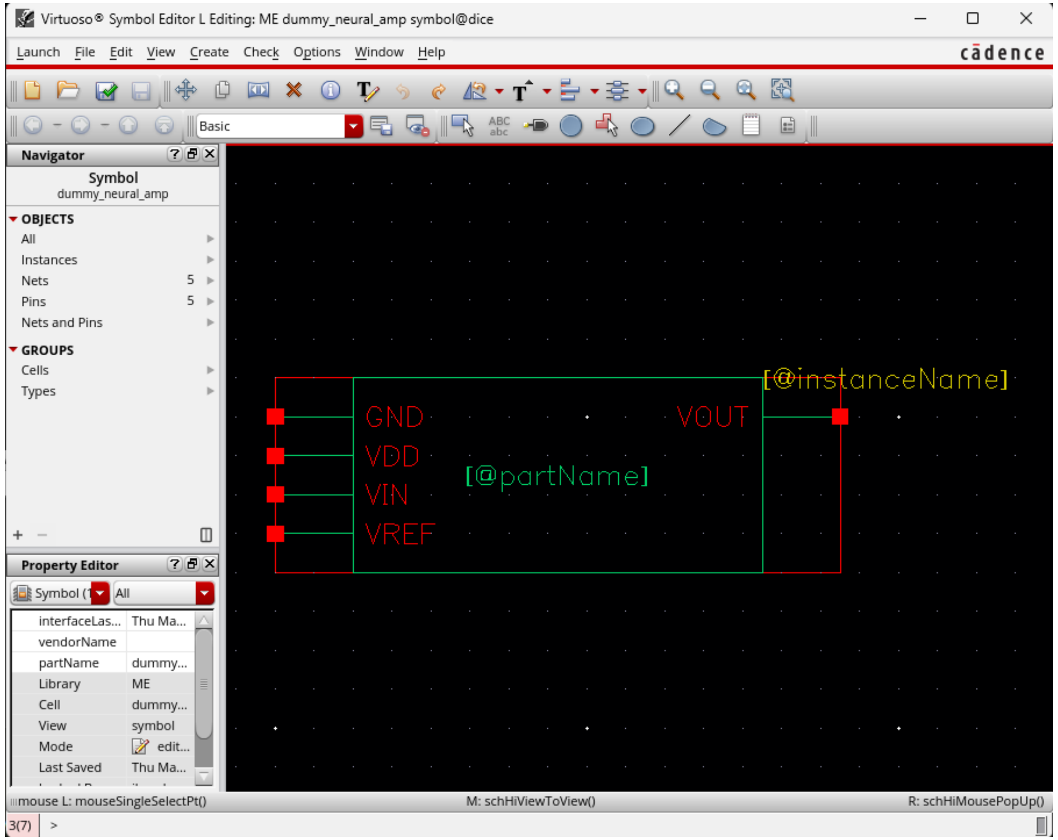
- Complete the schematic by placing all required ports using the Port tool shortcut P.
- Run Check & Save to verify the schematic and save the cellview.
- Generate the symbol by selecting Create → Cellview → From Cellview.
- Click all “ok”



Symbol View

❑Symbol Editing Flow

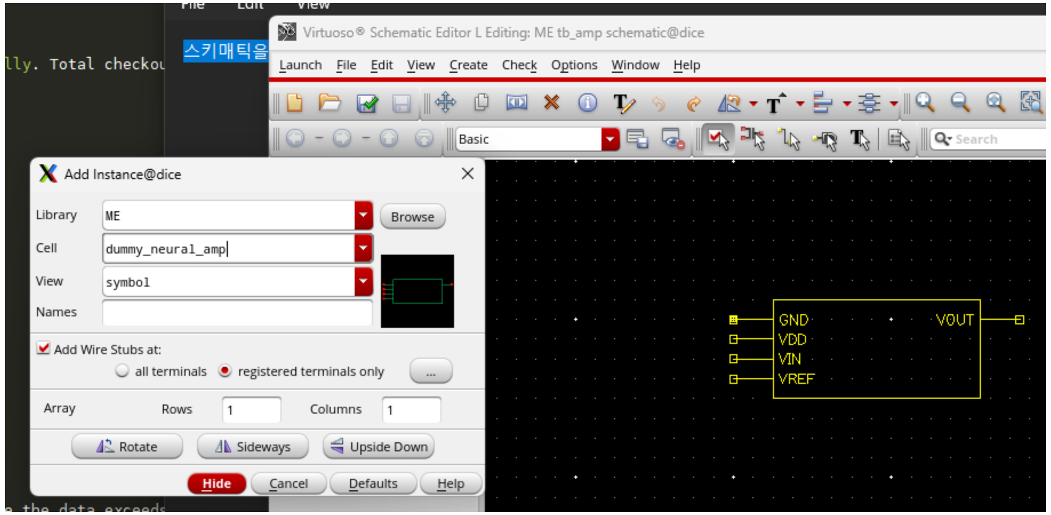
- Once the Symbol View opens, adjust the port locations as needed.
- Run Check & Save to verify and save the symbol.
- Close the Symbol View after saving.



Test Bench Schematic

❑Test Schematic Setup Flow

- Instantiate the newly created symbol in a test schematic.
- Complete the test schematic by adding the required input, output, and bias connections.
- Run Check & Save to verify and save the test schematic.
- Create a Maestro view for simulation setup and analysis.



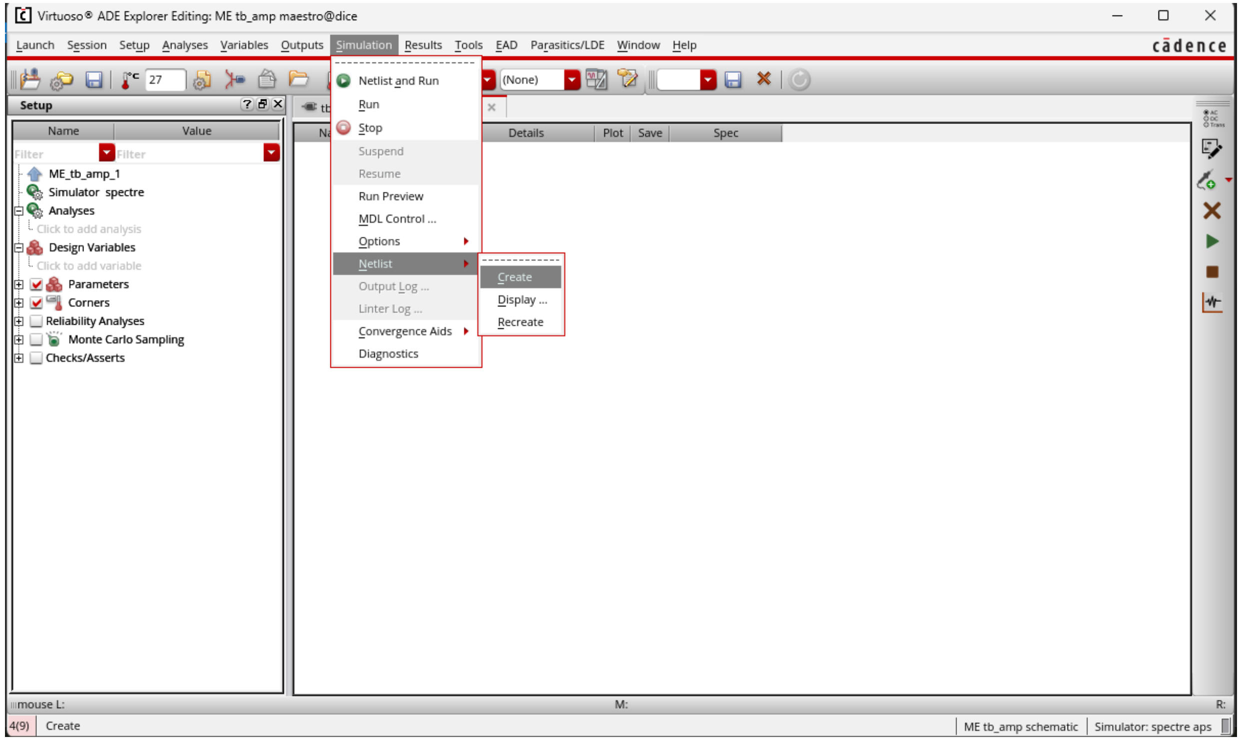
Netlist Creation

❑Netlist Generation in Maestro

- In Maestro, select Simulation → Netlist → Create.
- Generate the simulation netlist before running the analysis.

❑Replacing the Subcircuit Netlist

- In the generated netlist, copy the text from the .subckt line to the corresponding .ends line.
- Open dummy_neural_amp.scs in the amptest folder.
- Replace the existing subcircuit block with the copied netlist block while keeping the same overall structure.




```
simulator lang=spectre
ahdl_include "dummy_neural_amp.va"

subckt dummy_neural_amp GND VDD VIN VOUT VREF
    Q0 (VOUT VIN GND GND) npn_05v5_W1p00L1p00 area=1p m=10
    R0 (VDD VOUT) resistor r=500K
ends dummy_neural_amp
```

Configure.json setting

```
{
  "design_name": "veriloga_dummy_neural_amp",
  "work_dir": "run",
  "include_files": [],
  "library_sections": [
    {
      "path": "/home/eda/edk_cadence/sky130_release_0.1.0/models/sky130.lib.spice",
      "section": "tt"
    }
  ],
  "ahdl_include_files": [],
  "dut_netlist": "dummy_neural_amp.scs",
  "dut_subckt": "dummy_neural_amp",
  "dut_pins_order": ["VIN", "VREF", "VDD", "GND", "VOUT"],
  "spec": {
```

Matching with this text

This order should math with

```
simulator lang=spectre
ahdl_include "dummy_neural_amp.va"

subckt dummy_neural_amp GND VDD VIN VOUT VREF
    Q0 (VOUT VIN GND GND) npn_05v5_W1p00L1p00 area=1p m=10
    R0 (VDD VOUT) resistor r=500K
ends dummy_neural_amp
```

Run runtest.sh & check result in run folder ()

```
design: veriloga_dummy_neural_amp
area_total_p: 1670.2
static_current_a: 0.0007
power_dc_w: 0.0035
power_dynamic_w: 4.701371805999991e-05
power_score_basis_w: 0.0035
ac_nrmse_db: 0.303227
midband_gain_db: -25.190
lower_3db_hz: None
upper_3db_hz: None
atten_low_rel_mid_db: 1.4210854715202004e-14
atten_high_rel_mid_db: 1.4210854715202004e-14
tran_nrmse: 0.99945
vout_peak_to_peak_v: 0.00011
thd_db: -67.840
performance_nrmse_combined: 0.6513383590583701
```